Standard Form

Answer key · 11 marks total

Section A — Writing numbers in standard form

1. 8 µm = 0.000008 m (1)   = $\mathbf{8 \times 10^{-6}}$ **m** (1)

2. 30 nm = 0.00000003 m (1)   = $\mathbf{3 \times 10^{-8}}$ **m** (1)

3. 0.000004 m (1)   = $\mathbf{4 \times 10^{-6}}$ **m** (1)

Section B — Converting standard form back

4. 6×10^{-6} m = 0.000006 m (1)   = **6 µm** (1)

5. 1.5×10^{-4} m = 0.00015 m (1)   = **0.15 mm** (1)

6. **0.00009 m** (1)

Marking note: award method marks for correctly counting decimal places even if the final answer contains an arithmetic slip.